

MATH3303 – Lecture 9

Generators and Relations; Free Groups; Normal Closure

Review Notes

Reference: Lecture notes (supplementary; cf. Lee, Abstract Algebra)

Overview

This lecture introduces the language of *generators and relations*, which provides an efficient and compact way to describe groups without listing all elements explicitly:

- **Generating sets:** when a subset $S \subseteq G$ generates G .
- **Simple reflections** in Sym_n : the adjacent transpositions $(i \ i+1)$, and the claim that Sym_n is generated by them.
- **Words and relations:** formalising the notion of imposing relations among generators.
- **Free groups:** the group $F(S)$ of reduced words with no imposed relations, illustrated for $S = \{a\}$ and $S = \{a, b\}$.
- **Normal closure:** the smallest normal subgroup containing a given subset.

1 Generating Sets

Motivation

Many groups are too large to describe by listing explicit elements (e.g. $|\text{Sym}_n| = n!$). Generators and relations provide an efficient and compact way to describe groups in terms of basic elements and the rules they obey.

Definition 1.1: Generating set (from lecture)

Let G be a group. A subset $S \subseteq G$ is said to *generate* G if every element of G is a finite product of elements of S and their inverses.

We write $G = \langle S \rangle$.

Example 1.1: Generating sets of $\mathbb{Z}$ (from lecture)

For the additive group $(\mathbb{Z}, +)$, the following are generating sets:

$$\{1\}, \quad \{-1\}, \quad \{1, 3\}, \quad \{3, 5\}, \quad \{9, 10, 7\}.$$

The following are *not* generating sets:

$$\{2\}, \quad \{3\}, \quad \{2, 6\}, \quad \{4, 6\}.$$

(For example, $\{2\}$ fails because no combination of even integers reaches an odd integer.)

In $\mathbb{Z}_n$, the element a generates $\mathbb{Z}_n$ iff $\gcd(a, n) = 1$.

2 Simple Reflections in Sym_n

Definition 2.1: Simple reflections (from lecture)

In Sym_n (the symmetric group on n elements), for $i = 1, 2, \dots, n-1$, the *simple reflection* s_i is the transposition

$$s_i := (i \ i+1),$$

i.e. the permutation that swaps i and $i+1$ and keeps all other elements unchanged.

Theorem 2.1: Sym_n is generated by simple reflections (from lecture)

$\text{Sym}_n = \langle s_1, s_2, \dots, s_{n-1} \rangle$, i.e. Sym_n is generated by all simple reflections.

Proof. We show that every permutation can be expressed as a product of simple reflections, via a three-step reduction:

Step 1: Any permutation is a product of cycles.

This is a standard result: any $\sigma \in \text{Sym}_n$ can be written as a product of disjoint cycles.

Step 2: Any cycle is a product of transpositions.

A k -cycle $(i_1 \ i_2 \ \dots \ i_k)$ can be expressed as:

$$(i_1 \ i_2 \ \dots \ i_k) = (i_1 \ i_k)(i_1 \ i_{k-1}) \cdots (i_1 \ i_3)(i_1 \ i_2).$$

More generally, any transposition $(i \ j)$ with $i < j$ can be written as a product of transpositions of the form $(k \ \ell)$ with $k < \ell$.

Step 3: Any transposition is a product of simple reflections.

We show by induction on $j - i$ that $(i \ j)$ is a product of simple reflections.

- **Base case** $j = i + 1$: $(i \ j) = s_i$ is already a simple reflection.
- **Inductive step** $j > i + 1$: We use the identity

$$(i \ j) = (j-1 \ j) (i \ j-1) (j-1 \ j) = s_{j-1} (i \ j-1) s_{j-1}.$$

By the inductive hypothesis, $(i \ j-1)$ is a product of simple reflections. Hence so is $(i \ j)$.

Combining all three steps, every element of Sym_n is a product of simple reflections. □

3 Words and Relations

Definition 3.1: Word; relation (from lecture)

Let G be a group with generating set $S \subseteq G$. A *word* (in S) is a finite product of elements from S and their inverses.

A *relation* is a word that equals the identity element 1_G of G .

Example 3.1: Relations in $\mathbb{Z}_n$ (from lecture)

Let $G = \mathbb{Z}_n$ (additive notation). Let $a \in G$ with $\gcd(a, n) = 1$, so $G = \langle a \rangle$.

The relations include: $na = 0$, $2na = 0$, $3na = 0$, $\dots$ (and all multiples of na).

In general, $kna = 0$ is a relation for every $k \in \mathbb{Z}$.

Example 3.2: Relations in Sym_3 (from lecture)

Let $G = \text{Sym}_3$ and $S = \{s_1, s_2\}$ where $s_1 = (12)$ and $s_2 = (23)$.

The fundamental relations are:

$$s_1^2 = 1, \quad s_2^2 = 1, \quad (s_1 s_2)^3 = 1,$$

together with the relation $s_1 s_2 s_1 = s_2 s_1 s_2$ (the *braid relation*).

Imposing all of these relations gives the *presentation*:

$$\text{Sym}_3 = \langle s_1, s_2 \mid s_1^2 = 1, s_2^2 = 1, s_1 s_2 s_1 = s_2 s_1 s_2 \rangle.$$

4 The Free Group

Definition 4.1: Reduced word; free group $F(S)$ (from lecture)

Let S be a set. A word in S (and the formal inverses s^{-1} for $s \in S$) is called *reduced* if it contains no adjacent pair of the form ss^{-1} or $s^{-1}s$ for any $s \in S$.

The *free group on S* , denoted $F(S)$, is the set of all reduced words on S , with group operation defined by:

- (1) *Concatenation* of words, followed by
- (2) *Cancellation* of all adjacent pairs ss^{-1} or $s^{-1}s$ until the word is reduced.

The identity element is the *empty word* 1.

Example 4.1: $F(\{a\}) \cong \mathbb{Z}$ (from lecture)

Let $S = \{a\}$. Then $F(\{a\})$ consists of all reduced words in a and a^{-1} :

$$F(\{a\}) = \{\dots, a^{-2}, a^{-1}, 1, a, a^2, \dots\}.$$

This is the infinite cyclic group, so $F(\{a\}) \cong \mathbb{Z}$ (with $a \leftrightarrow 1 \in \mathbb{Z}$).

Example 4.2: $F(\{a, b\})$ is non-abelian (from lecture)

Let $S = \{a, b\}$. Then $F(\{a, b\})$ contains all reduced words such as:

$$1, \quad a, \quad b, \quad a^{-1}, \quad b^{-1}, \quad ab, \quad ba, \quad a^{-1}b, \quad ab^{-1}a, \quad \dots$$

Note $ab \neq ba$ in $F(\{a, b\})$, so the free group on two generators is non-abelian.

The word $aab^{-1}a = a^2b^{-1}a \in F(\{a, b\})$, since no cancellation is possible.

Remark 4.1: The universal property of free groups (from lecture)

The free group $F(S)$ imposes *no relations* beyond those required by the group axioms. Any group G generated by S is a *quotient* of $F(S)$: we have a surjective homomorphism $F(S) \twoheadrightarrow G$ whose kernel encodes exactly the relations satisfied in G .

This makes free groups the starting point for group *presentations*: a group G with generators S and relations R is written $G = \langle S \mid R \rangle$, meaning $G \cong F(S)/N$, where N is the smallest normal subgroup of $F(S)$ containing the set of relators R .

5 The Normal Closure

Lemma 5.1: Existence and uniqueness of the normal closure (from lecture)

Let G be a group and $S \subseteq G$. Then there exists a unique normal subgroup $\langle\langle S \rangle\rangle \trianglelefteq G$ such that:

- (1) $S \subseteq \langle\langle S \rangle\rangle$; and
- (2) if $N \trianglelefteq G$ and $S \subseteq N$, then $\langle\langle S \rangle\rangle \subseteq N$.

In other words, $\langle\langle S \rangle\rangle$ is the *smallest* (by inclusion) normal subgroup of G containing S . $\langle\langle S \rangle\rangle$ is called the *normal closure* of S in G .

Proof. Existence.

Let $\mathcal{X}$ be the collection of all normal subgroups of G that contain S :

$$\mathcal{X} = \{N \trianglelefteq G : S \subseteq N\}.$$

$\mathcal{X}$ is nonempty: $G \in \mathcal{X}$ since $S \subseteq G$ and $G \trianglelefteq G$.

Define

$$\langle\langle S \rangle\rangle := \bigcap_{N \in \mathcal{X}} N.$$

The intersection of normal subgroups is normal (Exercise: see Lee, Exercise 4.5), so $\langle\langle S \rangle\rangle \trianglelefteq G$.

By definition of intersection: $S \subseteq \langle\langle S \rangle\rangle$ (since $S \subseteq N$ for every $N \in \mathcal{X}$).

If $N \trianglelefteq G$ with $S \subseteq N$, then $N \in \mathcal{X}$, so $\langle\langle S \rangle\rangle \subseteq N$.

Uniqueness.

Suppose $P \trianglelefteq G$ also satisfies the two conditions. Then:

- Since P satisfies condition (2) (being the smallest such group), and $\langle\langle S \rangle\rangle \trianglelefteq G$ with $S \subseteq \langle\langle S \rangle\rangle$: $P \subseteq \langle\langle S \rangle\rangle$.
- Since $P \trianglelefteq G$ and $S \subseteq P$, we have $P \in \mathcal{X}$, so $\langle\langle S \rangle\rangle \subseteq P$.

Hence $P = \langle\langle S \rangle\rangle$. □

Remark 5.1: Connection to group presentations (from lecture)

In the context of free groups and presentations: if $G = \langle S \mid R \rangle$, the normal closure of R in $F(S)$ is the kernel of the quotient map $F(S) \twoheadrightarrow G$. That is,

$$G \cong F(S) / \langle\langle R \rangle\rangle.$$

The normal closure is needed (rather than just the subgroup generated by R) because the kernel of a group homomorphism is always a *normal* subgroup.

Checklist of Skills

- ☐ Define a generating set S of a group G ; write $G = \langle S \rangle$.
- ☐ Identify generating and non-generating sets of $\mathbb{Z}$; state the criterion $\gcd(a, n) = 1$ for a to generate $\mathbb{Z}_n$.
- ☐ Define simple reflections $s_i = (i \ i+1)$ in Sym_n .
- ☐ Prove that Sym_n is generated by simple reflections (three-step argument: any permutation is a product of cycles; any cycle is a product of transpositions; any transposition $(i \ j)$ is a product of simple reflections, by induction on $j - i$).
- ☐ Define a word and a relation in a group with generating set S .
- ☐ Write down relations for $\mathbb{Z}_n$ (multiples of na) and for Sym_3 (order relations and the braid relation).
- ☐ Define a reduced word and the free group $F(S)$ as the group of all reduced words under concatenation-then-cancellation.
- ☐ Verify $F(\{a\}) \cong \mathbb{Z}$.
- ☐ Give an example of a non-trivial non-reduced word in $F(\{a, b\})$ and reduce it.
- ☐ Explain why a group with generators S is a quotient of $F(S)$.
- ☐ State and prove the existence and uniqueness of the normal closure $\langle\langle S \rangle\rangle$ (via intersection of all normal subgroups containing S).
- ☐ Explain the role of the normal closure in group presentations: $G = \langle S \mid R \rangle \cong F(S)/\langle\langle R \rangle\rangle$.